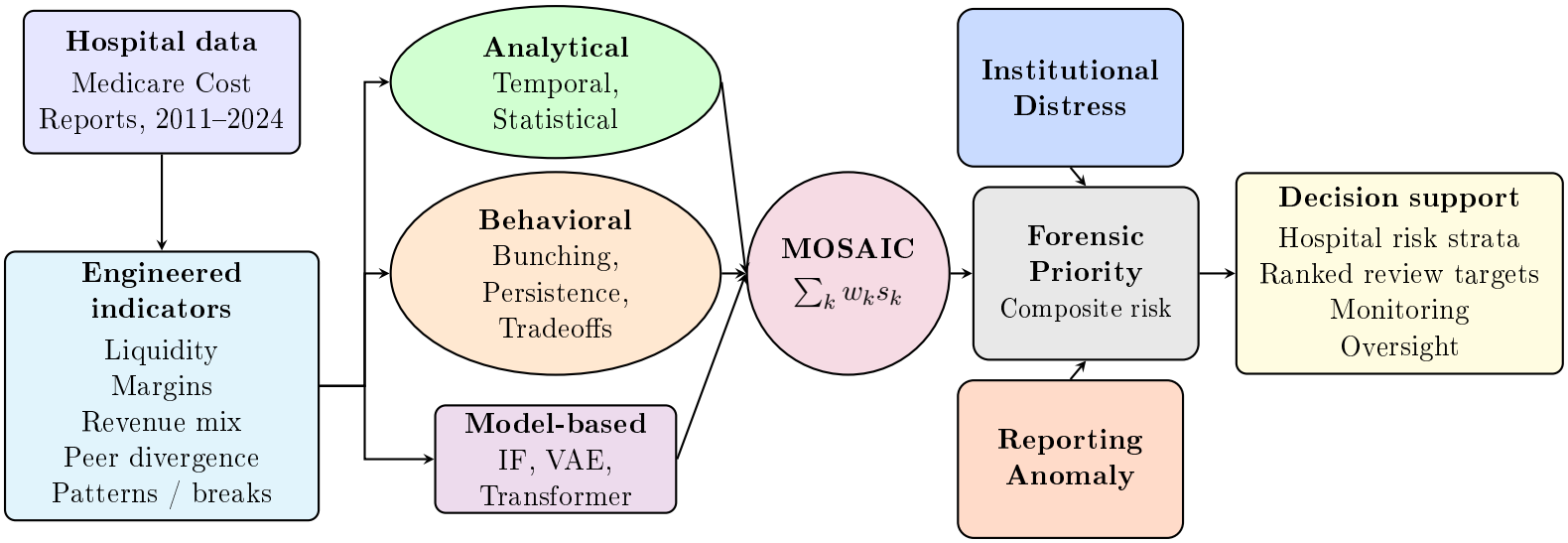
Graphical Abstract

MOSAIC: A Fusion Framework for Prospective, Forensic Anomaly Detection in Complex Administrative Panel Data



Highlights

MOSAIC: A Fusion Framework for Prospective, Forensic Anomaly Detection in Complex Administrative Panel Data

- Multi-objective fusion of analytical, behavioral, and model-based anomaly signals
- Baseline-trained weights (FY2012–2018) applied unchanged across all regimes
- Fully prospective evaluation using prior-period $(t-1)$ data to predict events in (t)
- Substantially higher concordance than unsupervised and reconstruction benchmarks
- Forward-looking identification of bankruptcies and enforcement actions above chance

MOSAIC: A Fusion Framework for Prospective, Forensic Anomaly Detection in Complex Administrative Panel Data

Abstract

Medicare hospital cost reports constitute one of the largest longitudinal administrative datasets describing the financial and operational behavior of U.S. hospitals, yet they remain underutilized for proactive oversight and audit prioritization. This study introduces MOSAIC (Multi-Objective Signal Aggregation for Interpretable Classification), a weakly supervised forensic prioritization framework integrating analytical, behavioral, and model-based anomaly signals through multi-objective optimization. Applied to 83,512 hospital facility-years from the Medicare HCRIS dataset (FY2011–2024), MOSAIC learns signal weights on a pre-pandemic Baseline regime (FY2012–2018) and applies them without modification to all other periods and validation data. MOSAIC achieves a Spearman concordance of $\rho_S = 0.5618$ with a financial distress anchor, compared with a maximum benchmark concordance of $\rho_S = 0.1173$. A signal-exclusion robustness analysis (MOSAIC-X) yields $\rho_S = 0.5394$, indicating that performance is not driven by anchor-adjacent variables or any single signal family. External validation using independent datasets of hospital bankruptcies and Medicare enforcement actions confirms that MOSAIC risk stratification anticipates real adverse events under a fully prospective design: risk scores constructed from prior-period $(t-1)$ data identify high-risk facilities in subsequent periods (t), with approximately 53–54% of events concentrated in the top two risk quintiles versus a 40% null expectation. These findings demonstrate that distress-aligned anomaly risk in administrative healthcare data is better captured through domain-constrained multi-signal integration than standalone unsupervised or high-capacity reconstruction models, and that MOSAIC provides a scalable and interpretable framework for data-driven audit prioritization.

Keywords: anomaly detection, weakly supervised learning, multi-objective optimization, signal fusion, audit prioritization

1. Introduction

The Medicare Hospital Cost Report Information System (HCRIS) is the most comprehensive longitudinal record of U.S. hospital financial activity, with annual filings from virtually all Medicare-participating facilities documenting revenues, expenses, capital structure, payer mix, and utilization (Centers for Medicare & Medicaid Services, 2023). Despite this richness, cost reports remain underutilized for prospective oversight: CMS and Medicare Administrative Contractors must prioritize limited investigative resources across tens of thousands of filings, yet audit selection relies largely on professional judgment and desk-review processes that may not systematically detect subtle or evolving anomalous reporting behavior (U.S. Department of Health and Human Services, Office of Inspector General, 2025). This gap is particularly consequential given the central role of cost reports in reimbursement and oversight within a system where Medicare accounts for roughly one-fifth of total U.S. health care spending (Centers for Medicare & Medicaid Services, 2024).

A useful conceptual lens for this problem distinguishes three forensic layers in hospital administrative data. The first is *institutional distress*: deterioration in underlying financial and operational condition manifested as margin compression, quality-measure decline, and organizational instability (Bazzoli et al., 2008; Joynt Maddox et al., 2019; Beauvais et al., 2026). The second is *reporting anomaly*: unusual patterns in submitted data including irregular line-item structure, digit-pattern deviations, and reconstruction failures under representation learning (Bolton and Hand, 2002; Chandola et al., 2009; Liu et al., 2008; Rousseeuw and Leroy, 2003). The third is their *joint signal space*: the presence of either or both signals within a facility-year, which defines the domain for forensic prioritization.[1] Figure 1 illustrates this framing. MOSAIC operationalizes this structure by integrating signals from both layers into a unified risk-stratification framework, without requiring their co-occurrence.

The signal fusion problem, how to combine heterogeneous anomaly in-

[1]This framing does not imply that distress-linked anomalies are fraudulent or that reporting anomalies arise from distress; it distinguishes conceptually among institutional state, reporting behavior, and their joint contribution to forensic risk.

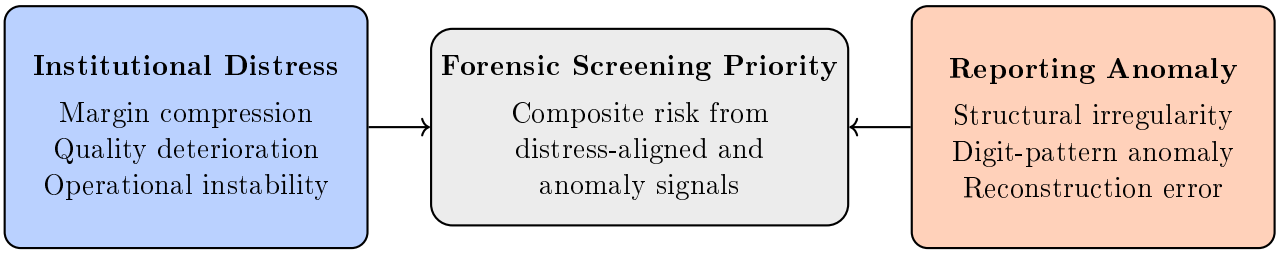


Figure 1: Conceptual framing of the signal domains combined by MOSAIC. Institutional distress and reporting anomaly are treated as distinct but complementary sources of forensic evidence whose signals are aggregated into a unified risk-prioritization framework.

dicators into a coherent composite risk score, remains underdeveloped in healthcare financial analytics. Existing work on hospital financial distress focuses on predictive or causal modeling rather than forensic detection (Bazzoli et al., 2008; Beauvais et al., 2026), while anomaly detection research in adjacent domains (Bolton and Hand, 2002; Chandola et al., 2009) rarely addresses the specific structure of administrative cost reports or the time-varying and cross-sectional heterogeneity in a national hospital panel. Most applied pipelines assign equal or heuristically specified weights to signals, embed assumptions that may fail across institutional contexts or time periods, and treat systematic missingness as noise rather than as a forensic signal (Marler and Arora, 2004; Keeney and Raiffa, 1993). Applications of anomaly detection to healthcare cost-report data specifically remain limited and typically employ single-method detection or heuristic fixed-weight fusion without formal weight identification (De Meulemeester et al., 2025).

This study develops MOSAIC (Multi-Objective Signal Aggregation for Interpretable Classification), a weakly supervised forensic risk-stratification framework for Medicare cost reports. MOSAIC is designed for prospective prediction and audit prioritization: it identifies facility-years that warrant attention, while causal attribution of the underlying anomaly remains a downstream task for the human auditor.

While the individual signals used in MOSAIC are constructed through unsupervised anomaly detection techniques, the aggregation weights are calibrated using a weak supervisory anchor representing institutional financial distress. This hybrid design preserves the exploratory advantages of unsupervised detection while resolving the indeterminacy of signal aggregation

through limited outcome-based calibration, and the learned weights are applied without modification to out-of-sample data across all holdout regimes.

The framework makes four contributions. First, it integrates heterogeneous anomaly indicators across analytical, behavioral, and model-based dimensions into a single composite score, allowing diverse signals to be evaluated within a unified structure. Second, it replaces heuristic fixed-weight fusion with a multi-objective optimization procedure that maximizes Spearman concordance with a financial distress anchor and cross-signal Kendall concordance, subject to directional monotonicity, regime smoothness, and hierarchical pooling constraints; the optimization is solved via GPU-accelerated L-BFGS (Nocedal and Wright, 2006) with multi-start initialization and simplex projection (Duchi et al., 2008), building on the identification framework of Bastian et al. (2017). Third, it demonstrates prospective external validity under a fully out-of-sample temporal design, with weights estimated on a pre-pandemic baseline and applied without modification to subsequent periods, and risk scores for year t constructed using only information from $t-1$ and evaluated against realized events in t. Fourth, it shows that domain-constrained, optimization-based signal fusion yields stronger identification and prioritization performance than established financial distress models and alternative unsupervised approaches under a common evaluation framework.

The remainder of this paper is organized as follows. Section 2 presents the MOSAIC framework. Section 3 reports the empirical analysis. Section 4 discusses implications and limitations. Section 5 concludes.

2. Methods

2.1. Notation

Tables 1 and 2 summarize the notation used throughout the paper. Indices, data variables, signal constructions, and model components are defined here for reference; complete variable construction details appear in Appendix B, and the full set-theoretic specification of the optimization problem is provided in Appendix D.

Table 1: Notation Reference I: Data, Signals, and Standardization

Notation Summary I

Indices

i — Hospital (facility) m — Performance measure t — Fiscal year c — Peer cohort r — Time regime k — Anomaly signal index ($k = 1, \ldots, K$; $K = 108$ in the baseline specification) j — SRRI constituent signal

Data and Anchor

X_{imt} — Raw or engineered value for facility i, measure m, period t
$s_m \in \{-1, +1\}$ — Direction indicator ($+1$ if higher X_{imt} is better)
$d_{it} \in [0, 1]$ — Financial distress anchor
$D_{it} = \mathbf{1}(d_{it} \geq 0.6)$ — Binary distress indicator
a_{it}^{CR}, a_{it}^{OM}, a_{it}^{DCOH} — Sigmoid proximity transforms

Signal Structure

$\mathcal{D} = \{\text{behavioral}, \text{temporal}, \text{statistical}, \text{model-based}\}$ — Signal dimensions
$\tilde{S}_{it}^{(k)} \in [0, 1]$ — Percentile-normalized anomaly signal (within cohort $\times$ regime)
$\mathbf{S}_{it} = (\tilde{S}_{it}^{(1)}, \ldots, \tilde{S}_{it}^{(K)})$ — Signal vector
$\tilde{S}_{it}^{(k)}$, $k \in \{\text{IF}, \text{VAE}, \text{TR}\}$ — Model-based signals (Isolation Forest, VAE, Transformer)

Standardization and Residuals

$Z_{\text{peer},imt}$ — Robust Z-score within peer cohort c
$Z_{\text{peer,signed},imt} = s_m \cdot Z_{\text{peer},imt}$ — Signed peer Z-score
$Z_{\text{peer}\times\text{yr},imt}$ — Robust Z-score within cohort $\times$ year cell
$Z_{\text{reg},imt}$ — Robust Z-score within cohort $\times$ regime cell
$Z_{\text{resid},imt}$ — Self-baseline residual

SRRI Components

$b_{j,it}$ — Percentile-ranked SRRI constituent signal
J — Number of SRRI signals ($J = 9$)
$b_{\text{persist},it}$ — Persistence component
$b_{\text{coord},it}$ — Coordinated-improvement component
$\text{SRRI}_{it} = \frac{1}{J} \sum_{j=1}^{J} b_{j,it}$ — Strategic Reporting Risk Index

See also Appendix B.

Table 2: Notation Reference II: Aggregation, Optimization, and Model Parameters

Notation Summary II

Weights and Aggregation

$w_k^{(c)}$ — Weight for signal k in cohort c (estimated on baseline, held constant across regimes)
$\mathbf{w}_{\text{global}}$ — Global pooled weight vector
$W^{(c)}$ — Kendall's concordance across signals (baseline observations, cohort c)
$F_{it} = \sum_k w_k^{(c(i))} \tilde{S}_{it}^{(k)}$ — MOSAIC fused score

Optimization

$\mathcal{L}(\mathbf{w})$ — MOSAIC objective function
n_c^{baseline} — Number of baseline observations in cohort c
n_0 — Prior sample size for shrinkage
$\alpha_c = \frac{n_c^{\text{baseline}}}{n_c^{\text{baseline}} + n_0}$ — Shrinkage weight
$\Delta^{(c)}(\mathbf{w}) = \mathbb{E}[F_{it} \mid d_{it} \geq 0.6] - \mathbb{E}[F_{it} \mid d_{it} \leq 0.3]$ — Separation statistic
Δ^{K-1} — Probability simplex
λ_p — Hierarchical pooling penalty

Moments and Parameters

$\mu_{m,c,r}$, $\sigma_{m,c,r}$ — Mean and SD for measure m, cohort c, regime r
μ_m^{nat}, σ_m^{nat} — National mean and SD
μ_c, Σ_c — Cohort mean vector and covariance matrix
$\bar{X}_{im,L}$, $\hat{\sigma}_{im,L}$ — Rolling mean and SD
$\sigma_{\text{group},cmt}$ — Cross-sectional SD

Thresholding and Uncertainty

τ — Threshold parameter
τ^* — Optimal threshold (FPR-constrained)
$\hat{\sigma}_{it}$ — Bootstrap uncertainty

Model Components

$\mu_\phi(\cdot)$, $\mu_\theta(\cdot)$ — VAE encoder and decoder mappings

See also Appendix D.

2.2. Data

The HCRIS panel was constructed from 14 consecutive annual CMS releases (HOSP10FY2011–HOSP10FY2024) (Centers for Medicare and Medicaid Services, 2026). Files exhibit inconsistent schemas, delimiters, and layout drift, requiring empirical schema inference and validation across all years. Approximately 40 numeric fields spanning seven domains were extracted, validated, and harmonized. Duplicate filings were resolved by retaining the most recent accepted report per provider-year. Facilities were assigned to five peer cohorts based on CMS ownership classification. Fiscal years were partitioned into four regimes: *Transition* (FY2011), *Baseline* (FY2012–2018), *COVID_Shock* (FY2019–2021), and *Recovery* (FY2022–2024), representing the transition from Medicare Hospital Cost Report Form 2552–1996 to 2552–2010, a stable pre-pandemic baseline period, pandemic-induced disruption, and post-pandemic normalization, respectively. All scaling parameters and decile cut-points were estimated on the baseline regime and applied unchanged thereafter to prevent temporal leakage. The final panel contains 83,512 facility-years with structural missingness preserved as an informative signal.

2.3. Framework Overview

MOSAIC is applied to the HCRIS panel described above. Figure 2 summarizes the six-layer pipeline: data ingestion and harmonization, preprocessing and leakage control, signal channel construction, multi-objective fusion, decision outputs, and external validation against independent adverse event datasets.

To provide context for the performance of the MOSAIC framework, we evaluate four alternative models. First, a legacy fixed-weight benchmark replicates traditional rule-based audit scoring approaches, assigning equal weight to the structural (VAE) and behavioral (SRRI) channels while the general-purpose IF receives half-weight as a domain-agnostic supplement. Second, a standalone Isolation Forest (IF) (Liu et al., 2008) trained on the same decile-encoded feature matrix (Section 2.4) provides a well-established density-based anomaly detection comparator. Third, a standalone Variational Autoencoder (Kingma and Welling, 2013) trained on identical inputs provides a reconstruction-based comparator and directly answers whether MOSAIC's performance is attributable to its structural channel alone. Fourth, a Transformer masked autoencoder (TabTransformerAE; $d_{\text{model}} = 64$, 2 layers, mask ratio = 0.15) trained on the same decile-encoded matrix provides

a contemporary deep learning comparator, adapting the masked autoencoding paradigm of He et al. (2022) to tabular inputs. All four baselines share identical underlying variables and training pools, ensuring that performance differences reflect model architecture and aggregation strategy rather than feature engineering or data access.

Two MOSAIC variants are evaluated alongside the primary specification. The signal-exclusion variant (MOSAIC-X) removes the two signal families most closely related to the distress anchor—operating-margin bunching and operating-margin quadratic trend signals, comprising 22 encoded signals and approximately 20% of baseline weight—and re-estimates the full optimization pipeline on the reduced signal set. MOSAIC-X assesses whether concordance depends on anchor-feature overlap or reflects the broader signal architecture. The financially augmented variant (MOSAIC-F) extends the signal set to include the three anchor variables themselves—current ratio, operating margin, and days cash on hand—after percentile-ranking and polarity correction, and serves as an upper-bound benchmark reflecting the performance attainable when direct financial information is incorporated. Neither variant alters the optimization procedure, temporal holdout design, or tier calibration method. See the appendices for a complete discussion of all variables and signals.

MOSAIC organizes anomaly indicators into three conceptual signal classes reflecting complementary perspectives on institutional behavior. Analytical signals capture deviations from expected relationships and distributional patterns in cost-report data, including peer-normalized residuals, dispersion-based anomalies, and statistical irregularities. Behavioral signals capture reporting patterns consistent with strategic responses to regulatory or financial incentives, including coordinated movements, persistence effects, and systematic non-reporting. Model-based signals derive from unsupervised and self-supervised learning algorithms, including IF, VAE, and Transformer-based reconstruction models, which detect nonlinear or high-dimensional anomalies not easily captured through analytical measures. This architecture allows MOSAIC to integrate interpretable rule-based diagnostics, behavioral pattern recognition, and learned anomaly representations into a unified framework for distress-aligned anomaly prioritization. See Appendix A and Appendix B for complete details of signals and variables.

External validity is assessed by linking pre-event MOSAIC and MOSAIC-F scores to two independent outcome datasets: hospital bankruptcy filings and Medicare enforcement actions occurring between 2022 and 2025. Event records are matched to HCRIS observations using a one-period year offset,

ensuring that all scores reflect information available prior to the recorded outcome. Risk tier assignments are calibrated on the baseline regime population and evaluated against a null expectation of 20% per tier, with enrichment in the High and Critical tiers constituting evidence of forward-looking discriminant validity. The external validation design and matching procedure are described in full in Section 2.10.

For comparison, established financial distress models including Ohlson (Ohlson, 1980), Zmijewski (Zmijewski, 1984), and BRKFSST (Beauvais et al., 2026) are evaluated under the same validation framework. Model scores are computed using published specifications and applied at the provider-year level without re-estimation, then aligned to outcome events using the same one-period offset to ensure that all evaluations are strictly prospective. Risk tiers for these benchmarks are calibrated using the same baseline population and null expectations as MOSAIC, enabling direct comparison of their ability to identify high-risk facility-years prior to realized adverse events. This parallel evaluation design ensures that observed performance differences reflect differences in signal structure and aggregation rather than differences in temporal alignment or calibration.

We now outline the components of the MOSAIC framework in detail. Section 2.4 describes the feature engineering and signal architecture used to construct analytical, behavioral, and model-based anomaly indicators. Section 2.5 introduces the financial distress anchor used for weak supervision. Section 2.6 presents the multi-objective programming formulation used to estimate signal weights. Section 2.7 defines the temporal holdout design. Section 2.8 introduces the signal-exclusion robustness variant (MOSAIC-X). Section 2.9 describes final score construction and thresholding. Section 2.10 details the prospective external validation design, and Section 2.11 presents the financial distress model benchmarks used for comparison.

2.4. Feature Engineering and Signal Architecture

Raw HCRIS variables are transformed into a high-dimensional signal space. Continuous variables are winsorized at the 1% and 99% tails to mitigate the influence of extreme reporting artifacts and data-entry errors that can distort downstream normalization and residual-based features. Because MOSAIC operates on relative, percentile-normalized, and pattern-based signals rather than raw magnitudes, this transformation preserves meaningful anomaly structure while stabilizing feature construction. Analytical features include peer-normalized scores and facility-specific residual deviations, see

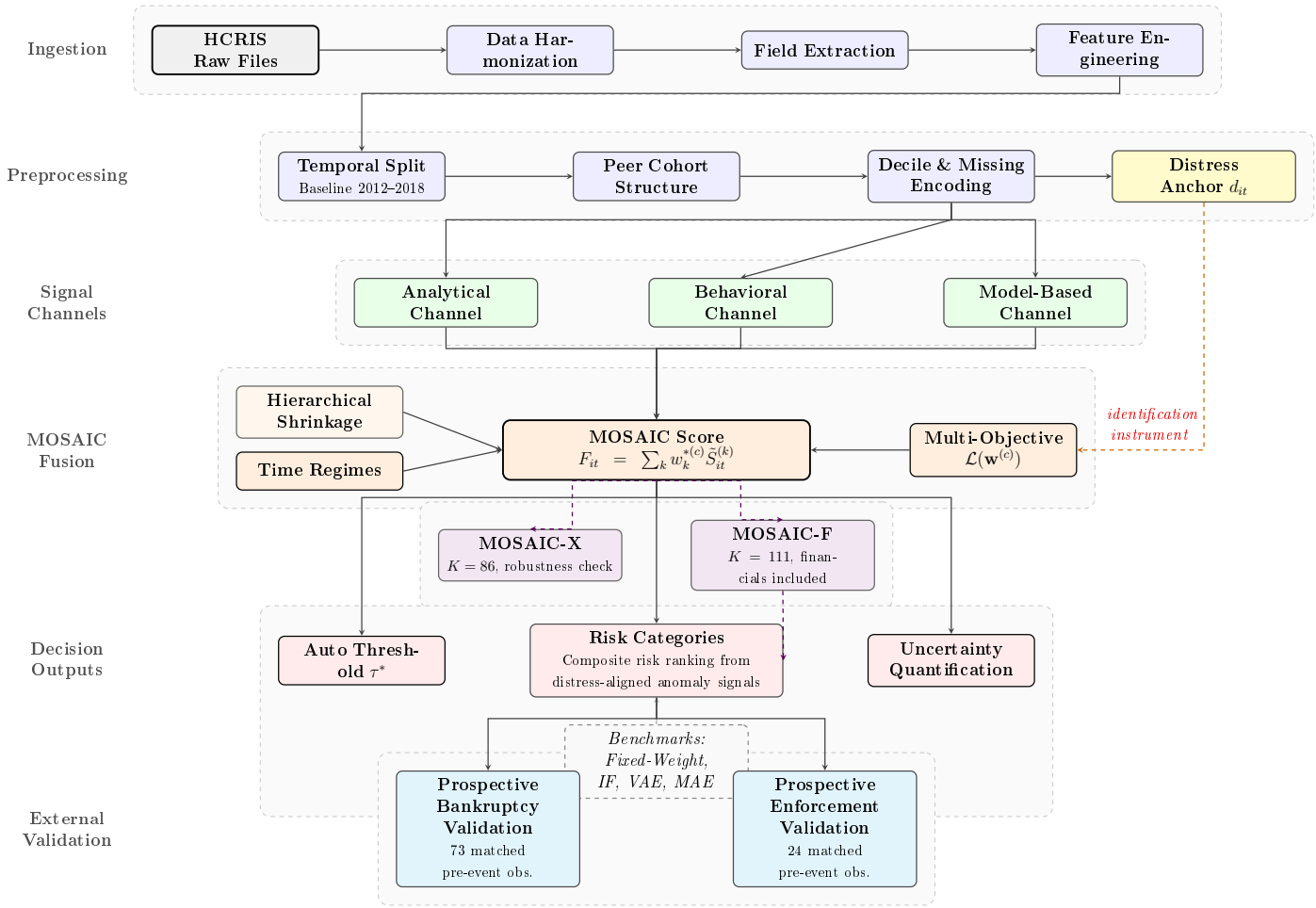


Figure 2: End-to-end MOSAIC pipeline across six layers: data ingestion, preprocessing, signal channel construction, multi-objective fusion with MOSAIC-X and MOSAIC-F variants, decision outputs, and external validation. MOSAIC-X denotes the signal-exclusion robustness variant ($K = 86$, anchor-adjacent signal families removed); MOSAIC-F denotes the financially augmented variant ($K = 111$, anchor variables included after percentile-ranking and polarity correction). Benchmarks include a fixed-weight composite, standalone Isolation Forest (IF), Variational Autoencoder (VAE), and Transformer masked autoencoder (MAE). Solid arrows indicate primary data flow; dashed orange arrows indicate the distress anchor as identification instrument; dashed violet arrows indicate variant specifications and their evaluation against independent outcome datasets.

Equation 1.

$$Z_{\text{resid},imt} = \frac{X_{imt} - \bar{X}_{im,8}}{\hat{\sigma}_{im,12}}. \tag{1}$$

Behavioral features capture persistence, coordination, and threshold-based reporting patterns. The Strategic Reporting Risk Index (SRRI) aggregates constituent percentile-ranked signals as

$$\text{SRRI}_{it} = 100 \cdot \Big(0.8 \cdot \frac{1}{J}\sum_{j=1}^{J} b_{j,it} + 0.2 \cdot \frac{b_{\text{persist},it} + b_{\text{coord},it}}{4}\Big), \tag{2}$$

where $b_{j,it} \in [0,1]$ denotes the percentile-ranked value of SRRI constituent signal j for facility i at time t, $J = 9$ is the number of base behavioral signals, $b_{\text{persist},it}$ captures temporal persistence in reporting patterns, and $b_{\text{coord},it}$ captures coordinated multi-measure movements consistent with strategic reporting behavior. The weighting scheme assigns greater emphasis to the cross-sectional distribution of behavioral signals (80%) while reserving a smaller share (20%) for dynamic components, reflecting their higher variance and sensitivity to short-term fluctuations. The division by 4 scales and attenuates the dynamic component to maintain its intended contribution relative to the averaged base signals. The 0.8/0.2 split is fixed *ex ante* to preserve interpretability and limit hyperparameter tuning within the behavioral sub-index; optimization of this split is deferred because the higher-level MOSAIC fusion layer already learns data-driven weights across signal families. Sensitivity analysis of the sub-index parameters confirms robustness: Spearman rank correlations between SRRI scores computed under the baseline parameterization $(0.8, 0.2, 4)$ and nine alternative specifications varying the static/dynamic split $\in \{0.7, 0.8, 0.9\}$ and scaling divisor $\in \{3, 4, 5\}$ range from 0.989 to 1.000, confirming that composite scores are insensitive to reasonable perturbations of these parameters.

Model-based detectors (IF, VAE, MAE) operate on an 11-bin decile-encoded feature matrix—10 ordinal bins plus an explicit missing-value bin—encoding non-reporting as an informative local signal. The MOSAIC fusion layer instead operates on continuous percentile-normalized signals derived from the same underlying variables, with missingness resolved prior to aggregation: missing values are imputed using Baseline-regime cohort medians (FY2012–2018, n = 43,529), after which non-reporting is recaptured through field-level missing-bin indicators entering the 108-signal catalog directly and

through the persistence component $b_{\text{persist},it}$ and coordinated-improvement component $b_{\text{coord},it}$ of the SRRI, which aggregate cross-variable, multi-period non-reporting into a single behavioral sub-index.

These differing treatments reflect architectural requirements rather than differences in available information: representation learning models require a fixed input matrix and therefore encode missingness locally and statically at the variable level, whereas MOSAIC models missingness as both a field-level event and a cross-variable temporal behavior. Accordingly, all models are evaluated on the same underlying data, but under representations appropriate to their respective modeling frameworks; enforcing a common encoding would either discard the higher-order temporal structure exploited by MOSAIC or violate the fixed-matrix input assumptions of the reconstruction models.

2.5. Financial Distress Anchor

MOSAIC employs a financial distress anchor as a weak supervisory signal. Let CR_{it} denote the current ratio, OM_{it} the operating margin, and $DCOH_{it}$ days cash on hand for facility i at time t. Each metric is transformed into a bounded proximity score using a logistic function centered at a domain-relevant threshold:

$$a_{it}^{CR} = \frac{1}{1 + \exp(5(CR_{it} - 1))}, \tag{3}$$

$$a_{it}^{OM} = \frac{1}{1 + \exp(30\, OM_{it})}, \tag{4}$$

$$a_{it}^{DCOH} = \frac{1}{1 + \exp(0.5(DCOH_{it} - 15))}. \tag{5}$$

The centering values reflect standard financial benchmarks: $CR_{it} = 1$ (liquidity break-even), $OM_{it} = 0$ (profit vs. loss), and $DCOH_{it} = 15$ (minimal liquidity buffer in days). The slope parameters $(5, 30, 0.5)$ control the sharpness of the logistic transition and are chosen to reflect differences in scale and dispersion across variables: operating margin is tightly concentrated around zero and uses a steep slope (30) to preserve sensitivity, days cash on hand spans a wider range and uses a gentler slope (0.5) to avoid saturation, and the current ratio lies between these extremes with an intermediate slope (5). These parameters are fixed *ex ante* and are not jointly

optimized with the signal weights; doing so would blur the boundary between the supervisory reference and the optimization target, potentially inducing circularity in the identification strategy. Sensitivity to anchor specification is assessed in Appendix C, where single-component anchor variants (CR-only, OM-only, DCOH-only) yield rank correlations with the full-anchor MOSAIC scores exceeding 0.94 in all cases, confirming that the framework's behavioral preferences are a property of the signal architecture rather than the specific anchor parameterization.

The composite distress anchor is defined as the average of the three proximity scores:

$$d_{it} = \frac{a_{it}^{CR} + a_{it}^{OM} + a_{it}^{DCOH}}{3}, \qquad d_{it} \in [0, 1]. \tag{6}$$

Facilities with $d_{it} \geq 0.6$ are classified as distressed. The anchor d_{it} is defined as the average of sigmoid-transformed financial proximity measures, so this threshold identifies facilities that lie on the adverse side of multiple financial benchmarks simultaneously. The cutoff is used to define a consistent high-risk partition for concordance in the optimization procedure; the anchor does not enter the signal set directly.

The distress anchor is intentionally parsimonious and is not intended to represent the full space of anomalous hospital behavior. Instead, it provides an interpretable external calibration signal centered on liquidity and operating stress, allowing MOSAIC to prioritize distress-aligned irregularities while limiting circularity with the anomaly feature set.

To preserve this separation, the anchor variables (current ratio, operating margin, and days cash on hand) are excluded from the anomaly signal catalog and used solely to construct d_{it}. The model must therefore identify independent combinations of engineered signals that are concordant with distress, rather than recovering it directly from the underlying variables.

A more subtle concern arises from signals that are correlated with financial performance but not algebraically derived from the anchor. This is mitigated in two ways. First, the optimization objective rewards concordance across the full distribution of facility-years, so signals with no independent anomaly information receive little weight. Second, signal-exclusion robustness checks are used to assess whether performance depends disproportionately on features closely related to the anchor variables; these results are presented in Section 3.

2.6. Mathematical Programming Formulation

For the complete mathematical programming formulation, see Appendix D. Let $\tilde{S}_{it}^{(k)} \in [0,1]$ denote normalized anomaly signals, constructed within cohort $\times$ regime cells so that higher values uniformly indicate greater anomaly severity. Let $w_k^{(c)}$ denote the weight assigned to signal k for cohort c, estimated using baseline observations and then held fixed across all regimes. The MOSAIC fused score is given by

$$F_{it} = \sum_{k=1}^{K} w_k^{(c(i))}\, \tilde{S}_{it}^{(k)}. \tag{7}$$

Although the signals are normalized within cohort $\times$ regime cells, the aggregation weights are estimated exclusively on the baseline regime and are not re-estimated for Transition, COVID-Shock, or Recovery periods. Accordingly, regime-specific variation enters the model through changes in the normalized signal values rather than through changes in the weight parameters.

For each cohort c, the weight vector $\mathbf{w}^{(c)} \in \Delta^{K-1}$ is obtained by solving, over baseline observations only,

$$\begin{aligned} \min_{\mathbf{w}^{(c)}} \mathcal{L}(\mathbf{w}^{(c)}) = &- \rho_S\big(F^{(c,\text{base})},\, d^{(c,\text{base})}\big) - \tfrac{1}{4}W^{(c)} \\ &+ 5\,\max\big(0, -\Delta^{(c)}(\mathbf{w}^{(c)})\big) \\ &+ \lambda_p \left\|\mathbf{w}^{(c)} - \mathbf{w}_{\text{global}}\right\|^2, \end{aligned} \tag{8}$$

subject to

$$\sum_{k=1}^{K} w_k^{(c)} = 1, \qquad w_k^{(c)} \geq 0 \quad (k = 1, \ldots, K). \tag{9}$$

Here, $W^{(c)}$ denotes Kendall's coefficient of concordance computed across signals over the baseline subset of cohort c. The separation statistic is defined over baseline observations as

$$\Delta^{(c)}(\mathbf{w}^{(c)}) = \bar{F}_{\mathcal{H}_{c,\text{base}}} - \bar{F}_{\mathcal{L}_{c,\text{base}}}, \tag{10}$$

where

$$\mathcal{H}_{c,\text{base}} = \{(i,t) : c(i) = c,\ d_{it} \geq 0.6,\ r(t) = \text{baseline}\}, \tag{11}$$

and

$$\mathcal{L}_{c,\text{base}} = \{(i,t) : c(i) = c,\ d_{it} \leq 0.3,\ r(t) = \text{baseline}\}. \tag{12}$$

The objective balances four components: concordance with the distress anchor, cross-signal agreement, monotonic separation between high- and low-distress observations, and hierarchical pooling toward a global weight vector. The relative magnitudes of these terms encode explicit trade-offs. The Spearman concordance term is unscaled and serves as the primary alignment objective. The Kendall concordance term is down-weighted (1/4) so that cross-signal agreement contributes to stability without dominating the ranking. The separation penalty is up-weighted (5) to strongly enforce monotonic ordering, reflecting its role as a structural constraint. The shrinkage parameter λ_p controls pooling toward the global weight vector, balancing cohort-specific adaptation against overfitting. The coefficients 1/4 and 5 were selected through empirical sensitivity analysis across the heterogeneous peer cohorts; robustness to perturbations of these values is documented in Appendix C.

2.7. Temporal Holdout Design

Model parameters and preprocessing components (including decile cutpoints, scaling parameters, and weight vectors) are estimated exclusively on the baseline regime (FY2012–2018) and applied unchanged to all other regimes, which constitute the holdout partition (Transition, COVID shock, and Recovery regimes). This ensures strict temporal separation, so performance in non-baseline regimes reflects genuine out-of-sample evaluation under distributional shift. All models are evaluated under the same regime definitions and baseline-derived calibration, ensuring comparability of temporal generalization despite differences in input representation.

2.8. Signal-Exclusion Robustness Variant (MOSAIC-X)

A central identification concern in weakly supervised frameworks that align signal aggregation with a financial reference is whether observed concordance reflects genuine anomaly detection or merely the recovery of information embedded in signals that share informational lineage with the reference itself (Ratner et al., 2017). Two signal families in the MOSAIC catalog are most susceptible to this concern: operating-margin bunching signals, which encode distributional compression around zero in the operating margin distribution, and operating-margin quadratic trend signals, which capture nonlinear trajectories in the same variable used as one of the three anchor components. Together these families comprise 22 encoded signals representing approximately 20% of total baseline weight in the full specification.

To assess the sensitivity of MOSAIC performance to these signals, a restricted variant designated MOSAIC-X was estimated by removing both families from the signal catalog prior to optimization, reducing the signal set from $K = 108$ to $K = 86$. With the exception of this exclusion, the full estimation pipeline was applied without modification: decile encoding, cohort- and regime-specific normalization, multi-objective optimization on the baseline regime, empirical Bayes shrinkage toward the global weight vector, and propagation of frozen weights to holdout regimes. Tier calibration and thresholding followed the same baseline-anchored procedure established in Section 2.7, with quintile thresholds estimated on the baseline regime and applied unchanged to all other regimes. MOSAIC-X therefore provides a conservative lower bound on performance under the assumption that the excluded signals contribute nothing beyond anchor-feature redundancy, and a concordance decline substantially smaller than their weight share would indicate that the framework's discriminative power is distributed across the broader signal architecture rather than concentrated in anchor-adjacent families. Full robustness diagnostics for MOSAIC-X are reported in Appendix C.

2.9. Final Scoring and Thresholding

The final MOSAIC score takes the form of Equation 7, with weights $\mathbf{w}^{*(c)}$ incorporating empirical Bayes shrinkage toward the global solution:

$$\mathbf{w}^{*(c)} = \alpha_c \, \hat{\mathbf{w}}^{(c)} + (1 - \alpha_c) \, \mathbf{w}_{\text{global}}, \qquad \alpha_c = \frac{n_c^{\text{baseline}}}{n_c^{\text{baseline}} + n_0}, \tag{13}$$

where n_c^{baseline} denotes the number of baseline observations in cohort c and n_0 is a prior sample size parameter controlling the strength of shrinkage toward the global weight vector (set to $n_0 = 50$ in all experiments, reflecting a prior equivalent to approximately one cohort-year of baseline observations). This formulation ensures that weight estimates for smaller cohorts are regularized toward the global structure, while larger cohorts rely primarily on their own estimated weights. These weights are estimated using baseline observations only and are not re-estimated for Transition, COVID-Shock, or Recovery regimes.

Facilities are flagged using a data-driven threshold τ^* (see Bastian et al. (2017)), defined as the solution to

$$\begin{aligned} \tau^* &= \arg\max_{\tau} \left[\mathbb{E}(F_{it} \mid D_{it} = 1) - \mathbb{E}(F_{it} \mid D_{it} = 0) \right] \\ &\quad \text{subject to } \Pr(F_{it} \geq \tau \mid D_{it} = 0) \leq \alpha, \end{aligned} \tag{14}$$

where $D_{it} = \mathbf{1}(d_{it} \geq 0.6)$ and $\alpha = 0.05$. The threshold 0.6 is set above the composite midpoint to require meaningful proximity to distress across the anchor components rather than marginal deviation; sensitivity to this choice is low given that the distress anchor enters only as a weak supervisory reference rather than a prediction target. The threshold τ^* maximizes the separation in expected scores between distressed and non-distressed facilities while constraining the false positive rate to α.

2.10. External Validation Design

To assess whether MOSAIC risk stratification anticipates real adverse events, scores were linked to two independent outcome datasets: hospital bankruptcy filings occurring between 2022 and 2025 (Becker's Hospital Review, 2025) and Medicare enforcement actions from the same period (Centers for Medicare & Medicaid Services, 2025a). These datasets are entirely external to the HCRIS panel used for model estimation, and all signal weights were estimated exclusively on the pre-pandemic baseline regime (FY2012–2018) and applied without modification to the recovery period observations used here — ensuring that the validation reflects both forward-looking signal detection and genuine temporal out-of-sample generalization. Event records were aligned to pre-event HCRIS observations by offsetting the event year by one period; for each fiscal year t, risk scores are constructed using information from $t-1$ and evaluated against outcomes in t, ensuring strictly prospective prediction with no contemporaneous information.

Matching used Medicare provider identifiers as the primary join key, supplemented by normalized facility name, city, and state matching for bankruptcy only. The fallback was withheld from enforcement records, where actions frequently target network or system entities rather than individual facilities. The resulting matched samples include 73 hospital-level bankruptcy observations and 24 hospital-level Medicare enforcement observations with valid pre-event HCRIS data.

Risk tier assignments were calibrated using quintile thresholds estimated on the baseline regime ($n = 43{,}529$), consistent with Section 2.7. Tiers are ordered by increasing risk and correspond to successive quintiles of the baseline score distribution: Low (0–20th percentile), Guarded (20–40th), Elevated (40–60th), High (60–80th), and Critical (80–100th). Under a null model with no discriminant power, each tier would contain 20% of events and the top two tiers jointly 40%. Enrichment above these benchmarks constitutes evidence of forward-looking discriminant validity. MOSAIC-F is

evaluated on the same matched samples and null benchmarks using the same baseline-calibrated thresholds, with anchor variables percentile-ranked and polarity-corrected prior to scoring to maintain the directional convention of the MOSAIC architecture.

2.11. Financial Distress Model Benchmarks

To assess whether MOSAIC's external discriminant validity exceeds that of established financial distress models, three published bankruptcy prediction models were evaluated against the same matched validation datasets described in Section 2.10. The models are Ohlson's O-score (Ohlson, 1980), Zmijewski's X-score (Zmijewski, 1984), and the hospital-specific BRKFSST model (Beauvais et al., 2026). Altman's Z-score (Altman, 1968) and its non-manufacturer variant (Altman, 2013) were excluded as both were developed on industrial firm samples and have not been validated for nonprofit or government hospital cost report data; the equity-based inputs they require are moreover unavailable or ill-defined for the predominantly nonprofit and government facilities in the HCRIS panel. Unlike MOSAIC, each of these models was estimated on historical bankruptcy samples with fixed published coefficients and is applied here in pure out-of-sample mode without re-estimation. Bankruptcy probabilities were computed for each facility-year using the original published coefficient vectors. For BRKFSST, variables requiring two-period lags restrict coverage to facility-years with at least three consecutive HCRIS filings. CMS Hospital Compare star ratings, required by the BRKFSST specification, are not available in HCRIS; the star rating indicator was set to zero for all facilities, which slightly overstates bankruptcy probability for the approximately 37% of Medicare hospitals holding a four- or five-star rating (Centers for Medicare & Medicaid Services, 2025b). Risk tiers were assigned using the same baseline-calibrated quintile thresholds applied to MOSAIC scores, permitting direct comparison under identical evaluation conditions and against the same 40% null expectation.

3. Results

3.1. Overall Model Performance

Table 3 presents the five-way comparison between MOSAIC and benchmark models across the full sample of 83,512 facility-year observations. Performance is evaluated using rank correlation with the distress reference (ρ_{global}), peer-year adjusted correlation ($\rho_{\text{peer}\times\text{yr}}$), and quartile separation.

MOSAIC delivers a substantial improvement over all benchmark approaches, achieving $\rho_{\text{global}} = 0.5618$ and $\rho_{\text{peer}\times\text{yr}} = 0.5796$ (both $p < 0.001$). These results represent a marked increase relative to standalone anomaly detection methods, including the Variational Autoencoder (VAE) ($\rho = 0.1173$) and IF ($\rho = 0.1104$), as well as the fixed-weight composite ($\rho = 0.0184$). In contrast, the Transformer-based masked autoencoder (MAE) exhibits negative correlation with distress ($\rho = -0.1189$), indicating that reconstruction error in this setting is misaligned with economically meaningful anomaly structure.

The magnitude of MOSAIC's improvement is substantial and statistically significant. Relative to the strongest global baseline (VAE), MOSAIC increases global correlation by $\Delta\rho = +0.4445$, while gains relative to IF and the fixed-weight composite exceed +0.44 and +0.54, respectively. Notably, IF and VAE exhibit near parity ($\Delta\rho = -0.0069$), confirming that no single-model baseline dominates and that the observed gains arise from structured aggregation rather than model selection.

Separation results further reinforce this conclusion. MOSAIC achieves a quartile separation of 0.4076, more than three times larger than either VAE (0.1156) or IF (0.1223), indicating substantially improved ranking of distressed versus non-distressed observations and sharper prioritization of high-risk facility-years.

Table 3: Five-Way Comparator — MOSAIC vs Baselines

Model	ρ_{global}	$\rho_{\text{peer}\times\text{yr}}$	Separation	N
Fixed-weight (0.4/0.4/0.2)	0.0184***	0.0401***	0.0507	83,512
Standalone VAE	0.1173***	0.1145***	0.1156	83,512
Standalone IF	0.1104***	0.1592***	0.1223	83,512
Transformer MAE	-0.1189***	-0.0626***	-0.0554	83,512
MOSAIC	**0.5618*** **	**0.5796*** **	**0.4076**	83,512

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. $\rho_{\text{peer}\times\text{yr}}$ controls for peer_group × source_year fixed effects. Separation is defined as $\mathbb{E}[\text{pct-rank} \mid Q4] - \mathbb{E}[\text{pct-rank} \mid Q1]$.

3.2. Baseline-Trained Weights and Generalization Across Regimes

A central design feature of MOSAIC is that aggregation weights are estimated exclusively on the baseline regime (FY2012–2018; $n = 43{,}529$) and then propagated unchanged to all other regimes. This design provides a direct

test of temporal generalization under distributional shift, as no re-estimation is performed outside the baseline period.

Despite this constraint, MOSAIC maintains strong performance across all regimes. Rank correlation remains high in Baseline ($\rho = 0.5991$), Transition ($\rho = 0.5663$), COVID Shock ($\rho = 0.5172$), and Recovery ($\rho = 0.5237$). These results indicate that the learned weighting structure captures persistent relationships between anomaly signals and distress, rather than regime-specific artifacts.

Importantly, MOSAIC consistently outperforms the best standalone model (IF) in every regime. The incremental lift (MOSAIC minus IF) remains large and stable, ranging from +0.3649 in the Transition period to +0.4475 in the Baseline regime. Even during the COVID Shock period, MOSAIC retains strong discriminatory power while simpler composite approaches deteriorate substantially.

Table 4: Regime-Stratified Performance (ρ_{global})

Regime	Fixed-wt	VAE	IF	MOSAIC	MOSAIC − IF
Baseline	0.0505	0.1322	0.1516	**0.5991**	0.4475
Transition	0.1652	0.1958	0.2013	**0.5663**	0.3649
COVID shock	-0.0254	0.0930	0.0917	**0.5172**	0.4255
Recovery	-0.0134	0.1418	0.1061	**0.5237**	0.4176

3.3. Comparative Model Behavior

The comparative results reveal three distinct patterns. First, single-model anomaly detectors (VAE and IF) exhibit modest but stable performance, indicating that each captures a partial view of distress-related structure but fails to fully align with the supervisory reference. Second, the fixed-weight composite performs poorly, suggesting that naive aggregation dilutes signal relevance and introduces noise. Third, the Transformer MAE fails to recover meaningful structure, with negative correlations implying that high-capacity reconstruction objectives capture dominant variation in the data rather than the distress-relevant deviations that matter for forensic prioritization. This result suggests that high-capacity reconstruction models may capture dominant variation in the data without isolating distress-relevant irregularities.

MOSAIC's performance gains therefore arise from adaptive weighting across heterogeneous signal classes, enabling the model to amplify informative signals while suppressing irrelevant or noisy components.

3.4. Decile Gradient and Ranking Behavior

Figure 3 illustrates the decile gradient of distress across model-implied rankings. MOSAIC produces a strictly increasing and strongly convex gradient, rising from 0.163 in the lowest decile to 0.620 in the highest decile, indicating precise ordering of observations by distress severity.

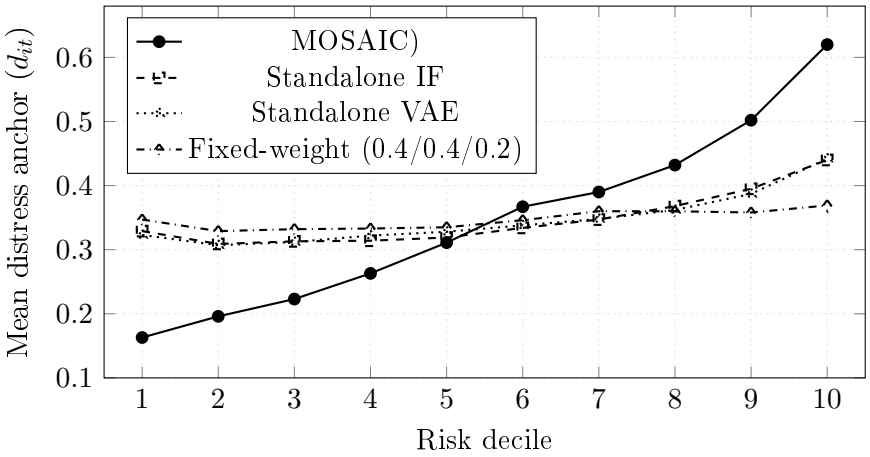


Figure 3: Decile gradient of mean distress anchor by model-implied risk decile. MOSAIC produces a strictly increasing and strongly convex gradient (0.163 to 0.620), while baseline models exhibit flat profiles with limited discrimination across the distribution.

In contrast, baseline models exhibit relatively flat gradients. IF increases only from 0.329 to 0.440, VAE from 0.323 to 0.442, and the fixed-weight composite from 0.347 to 0.369. These flatter profiles indicate limited discrimination across the distribution and weak separation of high-risk observations.

The steepness and monotonicity of the MOSAIC gradient confirm that the model produces economically meaningful rank ordering, which is critical for downstream prioritization and audit decision-making.

3.5. Signal Utilization and Weight Structure

MOSAIC integrates 108 signals spanning analytical, behavioral, and model-based dimensions, all with complete coverage across observations. The optimized weight distribution indicates that behavioral and temporal signals

dominate the aggregation, with the highest weights concentrated in measures capturing directional consistency in performance, clustering of values near reporting thresholds, divergence from peer trajectories over time, and decoupling between related financial indicators.

In contrast, model-based signals (including VAE- and Transformer-derived scores) receive minimal weight, indicating that representation learning contributes limited marginal information once richer behavioral and temporal signals are incorporated. This pattern suggests that distress is primarily reflected in dynamic and behavioral inconsistencies rather than static structural anomalies.

3.6. Robustness to Signal Exclusion

To assess robustness, we estimate a reduced model (MOSAIC-X) excluding 22 signals (approximately 20% of total weight mass). Despite this substantial reduction, performance remains strong: global correlation declines only modestly from 0.5618 to 0.5394 ($\Delta\rho = -0.0224$), with similar reductions in peer-year adjusted correlation. This result serves a dual purpose. First, it confirms that MOSAIC does not rely on a small set of dominant signals but instead benefits from redundant and distributed signal contributions. Second, and more directly, it provides evidence against overfitting the concordance objective: a model that had capitalized on sampling artifacts in the baseline period to maximize ρ_S would be expected to degrade sharply when 20% of its weight-bearing signals are removed. The modest observed decline—less than one-ninth of the removed weight share in concordance loss—indicates that the optimization has recovered a stable signal structure rather than fit noise. Ranking behavior and regime stability are similarly preserved across holdout periods, consistent with a weight solution that generalizes beyond the baseline sample on which it was estimated.

MOSAIC-X does not remove the anchor variables themselves, as these are not part of the anomaly signal set but serve only as an external supervisory reference. Instead, the robustness check targets signal families derived from similar underlying accounting relationships (e.g., operating-margin bunching and curvature) that could induce mechanical concordance. This design isolates potential feature–anchor overlap without eliminating the supervisory construct itself.

Importantly, ranking behavior and regime stability are preserved, indicating that MOSAIC does not rely on a small set of dominant signals but instead benefits from redundant and distributed signal contributions.

3.7. Interpretability of High-Risk Observations

To illustrate how MOSAIC scores decompose into interpretable components, Table 5 presents representative high-risk facility-years drawn from the upper tail of the distribution.

Across these observations, behavioral indicators consistently dominate the composite signal (approximately 57–59%), followed by analytical components (35–37%), with model-based signals contributing minimally (5–7%). This pattern aligns with the global weight structure and indicates that elevated risk is primarily driven by systematic reporting behavior rather than latent structural anomalies.

Scores represent ordinal rankings within cohort × regime cells and should be interpreted as relative prioritization rather than absolute anomaly magnitude. The decomposition provides a direct link between model output and underlying signal categories, supporting operational interpretability of flagged cases.

Table 5: Representative high-priority facility-years identified by MOSAIC (anonymized). Observations are selected from the upper tail of the risk distribution and correspond to confirmed high-risk cases. Dimensional percentages decompose each composite signal into behavioral, analytical, and model-based contributions.

Facility	Year	Behav.%	Analyt.%	Model%
Facility A	2020	57.8	35.9	6.3
Facility B	2022	57.1	36.8	6.1
Facility C	2019	58.0	36.1	5.9
Facility D	2024	58.0	35.6	6.4
Facility E	2024	57.1	36.8	6.1
Facility F	2019	58.8	35.2	5.9
Facility G	2016	58.2	36.0	5.9
Facility H	2023	58.6	35.5	5.8
Facility I	2020	57.4	36.1	6.5
Facility J	2022	57.4	36.9	5.7

3.8. External Validation of MOSAIC: Bankruptcy and Enforcement Outcomes

To evaluate external validity, MOSAIC outputs were linked to two independent outcome sets: hospital bankruptcy filings occurring between 2022 and 2025 (Becker's Hospital Review, 2025) and Medicare enforcement actions from the same period (Centers for Medicare & Medicaid Services, 2025a). In

both cases, events were aligned to pre-event HCRIS observations by offsetting the event year by one period. Critically, all signal weights were estimated exclusively on the pre-pandemic baseline regime (FY2012–2018) and applied without modification to the recovery period observations used here, ensuring that the validation reflects both forward-looking signal detection and genuine temporal out-of-sample generalization.

Matching was performed using Medicare provider identifiers, supplemented by limited name-based reconciliation. The resulting validation samples include 73 hospital-level bankruptcy observations and 24 hospital-level enforcement actions with valid pre-event data. Table 6 summarizes the distribution of these events across MOSAIC risk tiers.

Table 6: Distribution of adverse events across MOSAIC risk tiers. Expected proportions reflect the baseline population distribution under quintile partitioning (20% per tier; 40% for the top two tiers).

	Bankruptcy		Enforcement		Expected
Tier	Count	(%)	Count	(%)	(%)
Low	9	12.3	4	16.7	20.0
Guarded	9	12.3	4	16.7	20.0
Elevated	16	21.9	3	12.5	20.0
High	14	19.2	4	16.7	20.0
Critical	25	34.2	9	37.5	20.0
Top 40% (High+Critical)	39	53.4	13	54.2	40.0

3.8.1. Bankruptcy Validation

Bankrupt hospitals are concentrated in the upper tail of the MOSAIC risk distribution. Specifically, 53.4% of bankrupt facilities fall within the top two quintiles, compared to 40% expected under a null distribution (one-sided binomial test, $p = 0.014$). The highest-risk quintile alone contains 34.2% of bankrupt hospitals, representing a 71% increase over the expected 20% share (one-sided binomial test, $p = 0.003$).

These results indicate that MOSAIC identifies a subset of hospitals exhibiting pre-failure behavioral irregularities. The model does not capture all bankruptcies, which is expected given that financial failure may arise from exogenous shocks or structural factors not reflected in reporting behavior.

The observed enrichment therefore reflects selective identification of distress-associated anomalies rather than universal failure prediction.

3.8.2. Medicare Enforcement Validation

A similar pattern is observed for Medicare enforcement actions. Among matched hospital-level observations, 54.2% fall within the top two quintiles of the MOSAIC distribution, with 37.5% concentrated in the highest-risk quintile. These values exceed the expected 40% and 20% benchmarks, respectively. The matched sample of 24 observations is insufficient to achieve conventional statistical significance at the High+Critical level (one-sided binomial test, $p = 0.114$); however, the Critical-tier concentration is more compelling ($p = 0.036$).

Despite differences in outcome definition (bankruptcy representing institutional failure and enforcement actions reflecting regulatory intervention), the consistency of results across both datasets indicates that MOSAIC captures a common structure of institutional risk characterized by behavioral and temporal irregularities.

3.8.3. Cross-Outcome Consistency

The similarity in upper-tail concentration across bankruptcy and enforcement datasets is notable. In both cases, more than 53% of events occur within the top two quintiles of the MOSAIC distribution. This consistency provides evidence that the framework identifies persistent, cross-domain risk signals rather than overfitting to a specific outcome.

These findings support the interpretation of MOSAIC as a *distress-aligned anomaly prioritization framework* capable of identifying institutions exhibiting elevated risk profiles prior to observable adverse events. The alignment of results across independent outcome measures further indicates that behavioral anomaly signals provide early indications of institutional deterioration that are not fully captured by conventional financial indicators alone.

3.9. Comparison with Full Model Specification

To assess the incremental contribution of direct financial information, a full specification (MOSAIC-F) was estimated in which the anchor variables (current ratio, operating margin, and days cash on hand) were included in the signal set. As expected, this specification produces substantially higher concordance with the distress reference ($\rho_S = 0.8087$) and greater separation across the risk distribution. This reflects the direct inclusion of variables used

to construct the reference itself. However, the primary objective of MOSAIC is not to recover the distress anchor, but to identify independent behavioral and temporal signals that are concordant with it. The baseline specification achieves strong performance ($\rho_S \approx 0.56$) without direct access to these variables, indicating that a substantial portion of distress-aligned signal can be recovered from non-financial indicators alone. Accordingly, MOSAIC-F is best interpreted as an upper-bound benchmark, while the baseline specification provides a more informative measure of behavioral anomaly detection.

To evaluate whether the financial augmentation also improves external predictive validity, MOSAIC-F outputs were linked to the same bankruptcy and enforcement outcome sets used in Section 3 above, using an identical matching and year-offset procedure. Prior to scoring, the three financial ratio signals were percentile-ranked against the baseline population and polarity-corrected so that lower values of each ratio, reflecting greater financial stress, map to higher composite scores, consistent with the directional convention applied to all other signals in the MOSAIC architecture. The resulting validation samples include 73 hospital-level bankruptcy observations and 24 hospital-level enforcement actions with valid pre-event data. Table 7 summarizes the distribution of these events across MOSAIC-F risk tiers.

Table 7: Distribution of adverse events across MOSAIC-F risk tiers. Expected proportions reflect the baseline population distribution under quintile partitioning (20% per tier; 40% for the top two tiers).

	Bankruptcy		Enforcement		Expected
Tier	Count	(%)	Count	(%)	(%)
Low	10	13.7	3	12.5	20.0
Guarded	8	11.0	2	8.3	20.0
Elevated	18	24.7	5	20.8	20.0
High	17	23.3	6	25.0	20.0
Critical	20	27.4	8	33.3	20.0
Top 40% (High+Critical)	37	50.7	14	58.3	40.0

3.9.1. Bankruptcy Validation

MOSAIC-F reproduces the upper-tail enrichment observed for the baseline specification. Among matched bankruptcy observations, 50.7% fall within the top two quintiles compared to 40% under the null (one-sided binomial

test, $p = 0.042$), and 27.4% concentrate in the highest-risk quintile, a 37% increase over the expected 20% share ($p = 0.080$). The modest compression of the Critical tier relative to MOSAIC (27.4% vs. 34.2%) reflects a redistribution of score mass attributable to the inclusion of financial ratio signals, which smooth the composite distribution and compress the upper tail relative to the behavioral-only specification.

3.9.2. Medicare Enforcement Validation

As an upper-bound specification, MOSAIC-F shows higher enforcement enrichment than the baseline specification: 58.3% of matched enforcement observations fall within the top two quintiles ($p = 0.054$), exceeding both the 40% null and the 54.2% achieved by MOSAIC, with the Critical tier at 33.3% ($p = 0.089$) indicating near parity. Given the matched sample of 24 observations, these results should be interpreted directionally. The incremental lift over MOSAIC is concentrated in the High tier (25.0% vs. 16.7%), suggesting that financial signals contribute additional discriminant information for compliance risk beyond what behavioral and temporal channels capture alone.

3.9.3. Cross-Specification Comparison

Across both outcome types, MOSAIC and MOSAIC-F produce broadly comparable external validity, with MOSAIC exhibiting a slight advantage in bankruptcy discrimination (53.4% vs. 50.7%) and MOSAIC-F a meaningful advantage in enforcement detection (58.3% vs. 54.2%), consistent with financial signals carrying incremental compliance-specific information while contributing less beyond behavioral channels for bankruptcy identification. The two validation datasets share no common facilities over the 2022–2025 window, suggesting empirically distinct failure modes. Convergence of enrichment across two independent, non-overlapping outcome sets lends further support to the interpretation that MOSAIC's discriminant validity is a property of the signal architecture rather than a shared-outlier artifact. The theoretical intersection of financial distress and regulatory action remains an open empirical question, as temporal misalignment and sample size constraints preclude definitive assessment of co-occurrence within the current observation window.

3.10. Comparison with Established Financial Distress Models

Table 8 reports the external validation results for Ohlson's O-score (Ohlson, 1980), Zmijewski's X-score (Zmijewski, 1984), and the hospital-specific BRK-

FSST model (Beauvais et al., 2026), evaluated against the same matched bankruptcy and enforcement samples used previously. All three models are applied with fixed published coefficients in pure out-of-sample mode. MOSAIC and MOSAIC-F results are reproduced for reference.

Table 8: External validation comparison across MOSAIC specifications and established financial distress models. High+Critical (H+C) reflects the proportion of matched adverse events in the top two risk quintiles; Critical (Crit) reflects the proportion in the highest quintile alone; null expectations are 40% and 20% respectively. One-sided binomial p-values reported for the H+C comparison against the 40% null; n.s. denotes $p > 0.10$.

	Bankruptcy ($n = 73$–79)			Enforcement ($n = 24$)		
Model	H+C (%)	Crit (%)	p	H+C (%)	Crit (%)	p
MOSAIC	53.4	34.2	0.014	54.2	37.5	n.s.
MOSAIC-F	50.7	27.4	0.042	58.3	33.3	n.s.
Zmijewski X-score	43.0	29.1	n.s.	41.7	25.0	n.s.
Ohlson O-score	39.2	24.1	n.s.	37.5	20.8	n.s.
BRKFSST	30.4	11.4	n.s.	50.0	20.8	n.s.
Null expectation	40.0	20.0	—	40.0	20.0	—

Bankruptcy n differs across specifications: MOSAIC results are based on $n = 73$ matched observations; BRKFSST requires two-period lags, restricting coverage to facility-years with at least three consecutive HCRIS filings ($n = 79$ eligible observations prior to event matching). CMS Hospital Compare star ratings unavailable in HCRIS; BRKFSST star rating term set to zero for all facilities (Centers for Medicare & Medicaid Services, 2025b), which may attenuate discrimination. All models applied with fixed published coefficients without re-estimation.

MOSAIC is the only model to achieve statistically significant enrichment above the null for either outcome, with 53.4% of bankruptcy observations and 54.2% of enforcement observations falling in the top two risk quintiles ($p = 0.014$ and $p = 0.114$ respectively). Among the financial distress models, Zmijewski's X-score produces the strongest bankruptcy discrimination at 43.0%, marginally above the null but not statistically distinguishable from it ($p = 0.329$). Ohlson's O-score falls at the null on bankruptcy (39.2%, $p = 0.597$) and below it on enforcement (37.5%). Both results are consistent with the known tendency of financial ratio models estimated on corporate samples to generalize poorly to the hospital sector, where ownership structure, payer mix, and cost report accounting conventions differ substantially

from the manufacturing and retail firms that dominated Ohlson and Zmijewski's estimation samples.

BRKFSST presents a striking asymmetry: 30.4% High + Critical on bankruptcy — below the null and the weakest result across all models — against 50.0% on enforcement, the second highest enforcement result after MOSAIC-F. The bankruptcy underperformance likely reflects the unavailability of CMS star ratings in HCRIS, which forces the largest coefficient in the BRKFSST specification to zero and removes its primary discriminating variable. The enforcement result suggests that BRKFSST's operational variables — accounts receivable ratios, bed utilization, and labor compensation — carry compliance-relevant information that is partially complementary to MOSAIC's behavioral signal architecture, consistent with the broader finding that enforcement risk has a distinct financial signature from the failure pathway leading to bankruptcy.

These results indicate that MOSAIC's external discriminant validity is not attributable to financial ratio recovery. The financial distress models, which operate exclusively on balance sheet and income statement ratios, fail to replicate MOSAIC's enrichment on either outcome despite having access to the same financial variables that underlie the distress anchor used to calibrate MOSAIC's weights. The behavioral and temporal signal architecture therefore contributes discriminant information that is not recoverable from financial structure alone.

4. Discussion

4.1. Identification Strategy and Its Scope

MOSAIC employs a weakly supervised identification strategy in which a composite of financial indicators (current ratio, operating margin, and days cash on hand) serves as an external reference for resolving indeterminacy in signal aggregation. Rather than acting as a prediction target, this reference enters the objective as an alignment criterion, selecting weight vectors that produce rankings consistent with financial distress. Identification is further supported by a temporal holdout design: all parameters are estimated on the baseline regime (FY2012–2018) and applied unchanged to all other regimes — Transition, COVID-Shock, and Recovery — with concordance remaining above $\rho_S = 0.50$ across all holdout periods including the COVID-shock years. A signal-exclusion robustness check yields only a modest decline in concordance ($\rho_S = 0.5618 \rightarrow 0.5394$), confirming that performance reflects

a broadly distributed signal architecture rather than anchor–feature overlap. Accordingly, MOSAIC identifies facility-years that are both anomalous and aligned with financial distress — it is not a direct fraud detector but a distress-aligned anomaly prioritization framework, and facilities that are anomalous without financial stress, or distressed without reporting irregularities, may not be captured.

4.2. Behavioral Versus Structural Signals

The optimized weight structure is dominated by behavioral and temporal signals while structural components — particularly VAE reconstruction error — receive negligible weight, and the Transformer masked autoencoder yields negative concordance ($\rho_S = -0.1189$), confirming that reconstruction-based objectives capture dominant variation in the data rather than rare or policy-relevant deviations. Distress-aligned anomalies are therefore expressed primarily through behavioral and temporal inconsistencies rather than static structural deviations. The external validation results further illuminate the differential role of financial signals: MOSAIC-F produces broadly comparable bankruptcy discrimination to the baseline specification (50.7% vs. 53.4%) while achieving meaningful incremental lift for enforcement detection (58.3% vs. 54.2%), suggesting that behavioral signals drive pre-failure identification while financial signals capture compliance-specific risk where operational stress and reporting irregularities co-occur more proximately. Established financial distress models—Ohlson, Zmijewski, and BRKFSST—fail to achieve statistically significant enrichment on either outcome under these evaluation conditions, a result consistent with known limitations of models estimated on corporate manufacturing samples (Ohlson, 1980; Zmijewski, 1984) and with the unavailability of a key discriminating variable in the hospital-specific specification (Beauvais et al., 2026), suggesting that discriminant validity derives primarily from the behavioral signal architecture rather than financial ratio recovery alone, and that MOSAIC-F is best reserved for enforcement and compliance screening rather than general oversight.

4.3. Limitations

Several limitations warrant consideration. The framework relies on a financial proxy reference rather than confirmed irregularities, so MOSAIC identifies distress-aligned anomalies that may not capture operationally or strategically anomalous but financially neutral behavior; direct validation against confirmed audit findings remains outstanding. The temporal regime

structure is researcher-defined, and future disruptions may exhibit different characteristics requiring recalibration. A single reference across heterogeneous hospital types may mask cohort-specific dynamics. Finally, the signal catalog is tied to HCRIS and CMS quality measures as of fiscal year 2024 and will require periodic updates as reporting standards and reimbursement policies evolve.

4.4. Future Directions

Several extensions of the MOSAIC framework merit further investigation. Linking pre-event scores to confirmed audit outcomes — such as Recovery Audit Contractor determinations, OIG exclusion actions, or False Claims Act settlements — would enable supervised refinement of the reference and direct calibration against adjudicated irregularities. Adaptive optimization strategies tailored to cohort-regime characteristics may improve performance in small or volatile samples, and dynamic peer cohort assignment based on evolving institutional characteristics could sharpen peer-relative comparisons. Extension to other administrative healthcare datasets — including skilled nursing facility cost reports, where initial application suggests broad portability — and integration of higher-frequency data sources would further enhance generalizability and enable earlier detection of institutional deterioration.

5. Conclusion

This study developed and evaluated MOSAIC, a weakly supervised anomaly prioritization framework for Medicare hospital cost reports, building on the broader literature on anomaly detection and fraud-screening systems in complex administrative environments (Bolton and Hand, 2002; Chandola et al., 2009; Aggarwal, 2017). Applied to 83,512 facility-years spanning fiscal years 2011–2024 across five peer cohorts (government, nonprofit, for-profit, other, and critical access hospital facilities), MOSAIC replaces arbitrary fixed-weight signal fusion with a multi-objective optimization procedure that identifies aggregation weights consistent with distress-aligned concordance, cross-signal agreement, directional monotonicity, temporal stability, and hierarchical pooling, without requiring labeled outcome data.

Five findings stand out. Aggregation weights are a first-order modeling decision: MOSAIC achieves $\rho_S = 0.5618$ ($p < 0.001$, $n = 83{,}512$), substantially outperforming standalone models and the fixed-weight composite,

which yields near-zero concordance ($\rho_S = +0.018$). The strong monotone gradient in MOSAIC rankings indicates systematic alignment with distress, underscoring the central role of data-driven weight identification when labeled outcomes are unavailable. Weights estimated on the pre-pandemic Baseline regime (FY2012–2018) generalize to all holdout periods including the COVID-shock years, with concordance remaining above $\rho_S = 0.50$ throughout, suggesting that learned behavioral patterns reflect persistent institutional risk structure rather than regime-specific artifacts.

Behavioral and temporal signals dominate structural features. Quality deterioration and operating-margin bunching account for a large share of the weight mass, while reconstruction-based signals contribute minimally. Excluding the signal families most closely related to the distress anchor reduces concordance only modestly ($0.5618 \rightarrow 0.5394$), and a Transformer masked autoencoder yields negative concordance ($\rho_S = -0.1189$), indicating that reconstruction-based objectives capture dominant variation in the data rather than rare or policy-relevant deviations. Together these results suggest that distress-aligned anomaly prioritization is better served by domain-constrained signal fusion than by increasing model capacity.

External validation against independent outcome sets provides preliminary evidence of forward-looking discriminant validity. Among hospital bankruptcy filings and Medicare enforcement actions matched to pre-event HCRIS observations one fiscal year prior to the recorded outcome, approximately 53–54% of events fall within the top two MOSAIC risk quintiles against a 40% null expectation, with the highest-risk quintile alone accounting for 34% of bankrupt facilities and 37.5% of enforcement actions. A financially augmented specification (MOSAIC-F) shows incremental lift for enforcement detection (58.3%) consistent with financial signals carrying additional compliance-specific information, though modest matched sample sizes warrant caution in generalizing this asymmetry.

Established financial distress models — Ohlson's O-score, Zmijewski's X-score, and the hospital-specific BRKFSST model — applied with fixed published coefficients in pure out-of-sample mode fail to achieve statistically significant enrichment on either outcome, despite operating on the same financial variables underlying the MOSAIC distress anchor. This confirms that MOSAIC's discriminant validity derives from its behavioral and temporal signal architecture rather than financial ratio recovery alone.

All inputs are derived from publicly available HCRIS data, and the resulting stratification provides a practical triage mechanism with each flagged

facility-year decomposable into interpretable signal dimensions (Turban et al., 2005; Shim et al., 2002; Chen et al., 2012). Future work should prioritize linking MOSAIC scores to confirmed audit outcomes — such as Recovery Audit Contractor determinations or False Claims Act settlements — to enable supervised refinement of the reference and direct calibration against adjudicated irregularities. Extension of the framework to additional administrative datasets would support broader deployment and the development of unified anomaly detection systems across healthcare programs.

Disclosures

Competing interests. The authors declare that they have no known financial or non-financial competing interests that could have influenced the work reported in this paper.

Funding sources. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Author contributions. All authors contributed equally to the conceptualization, methodology, software development, analysis, visualization, and writing of this manuscript.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process. During the preparation of this work, the authors used OpenAI's ChatGPT to assist with drafting and editing of prose for clarity and readability, and Anthropic's Claude Sonnet 4.6 to assist with programming tasks, including code generation, refinement, and debugging related to data generation, analysis, and visualization. After using these tools, the authors reviewed and edited all content as needed and take full responsibility for the content of the published article.

Ethics approval and consent to participate. This study did not involve human participants, human subjects data, or animal experimentation and therefore did not require institutional ethical approval or informed consent.

Data and code availability. All datasets used in this study are publicly available. These data may be downloaded at the following link. `https://www.cms.gov/data-research/statistics-trends-and-reports/cost-reports/cost-reports-fiscal-year`. All code is available online at `https://github.com/dustoff06/CMS`.

Appendix A. Signal Catalog

Table A.9 summarizes representative signals from the full MOSAIC catalog of 108 encoded features. Signals are organized into three channels: analytical, behavioral, and model-based. The complete catalog, including decile cut-points for all partially observed signals, is provided in the supplementary data archive. Coverage rates are computed over the full panel (FY2011–2024) and reflect the fraction of facility-years with observed (non-missing) values. The "higher = worse" column indicates whether each signal is used in its natural direction or sign-reversed prior to percentile normalization.

Anchor variables (current ratio, operating margin, and days cash on hand) are not included in the signal catalog and are used exclusively to construct the supervisory reference.

Table A.9: MOSAIC signal catalog (selected signals; full catalog available in supplementary archive).

Plain name	**Identifier**	**Channel**	**Higher = worse**	**Coverage**
Analytical channel				
Peer divergence trajectory	`peer_divergence_year`	Analytical	Yes	99.2%
z-score trend	`z_delta_mean`	Analytical	Yes	94.4%
Operating-margin quadratic trend	`margin_curv6`	Analytical	Yes	63.5%
Rank instability	`rank_instability_mean`	Analytical	Yes	90.3%
Decoupling index	`decoupling_index`	Analytical	Yes	87.5%
Residual volatility	`z_resid_roll_std4`	Analytical	Yes	56.5%
Mahalanobis distance	`mahal_relational_score`	Analytical	Yes	100.0%
Digit anomaly score	`benford_anomaly_score`	Analytical	Yes	37.9%
Behavioral channel				
Quality deterioration composite	`goodz_weighted_sum`	Behavioral	Yes	94.4%
Margin bunching signal	`bunch_signal_operating_margin`	Behavioral	Yes	94.6%
Strategic reporting risk index	`SRRI`	Behavioral	Yes	100.0%
Coordinated improvement	`pct_measures_improving`	Behavioral	Yes	100.0%
Persistence index	`tslf`	Behavioral	Yes	100.0%
Tradeoff signal	`tradeoff_signal`	Behavioral	Yes	100.0%
Model-based channel				
Cost-report reconstruction error	`vae_gaming_score`	Model-based	Yes	100.0%
Isolation Forest anomaly score	`if_pct_decile`	Model-based	Yes	100.0%
Transformer reconstruction error	`transformer_pct`	Model-based	Yes	100.0%

Appendix B. Variable Construction Reference

This appendix provides a complete reference for the variable construction pipeline described in Section 2.2. Variables are organized into three signal channels: analytical, behavioral, and model-based. Tables B.10–B.14 document the engineered financial indicators, derived analytical statistics, behavioral anomaly features, model-based signals, and partial-coverage variables used in the MOSAIC framework. Directional conventions, formula definitions, and panel coverage rates are reported where applicable.

All signals are transformed into percentile ranks within cohort $\times$ regime cells, with cut-points estimated on the Baseline regime and applied uniformly across all periods. Missing values are encoded as an explicit category within the decile framework, allowing non-reporting behavior to function as an informative signal.

Tables 1 and 2 provide the complete notation. The variables used to construct the supervisory reference (current ratio, operating margin, and days cash on hand) are included below for completeness but are not part of the anomaly signal set.

Table B.10: Engineered Financial Indicators Derived from HCRIS Fields

Category	Term	Formula
Operational Efficiency	Revenue per Patient Day (RevPerDay_{it})	$\frac{\text{Net Patient Revenue}_{it}}{\text{Patient Days}_{it}}$
Operational Efficiency	Expense per Patient Day (ExpPerDay_{it})	$\frac{\text{Total Operating Expenses}_{it}}{\text{Patient Days}_{it}}$
Liquidity	Days Cash on Hand (DCOH_{it})	$\frac{\text{Cash \& Investments}_{it}}{\text{Total Operating Expenses}_{it}/365}$
Revenue Integrity	Cost-to-Charge Ratio (CCR_{it})	$\frac{\text{Total Operating Expenses}_{it}}{\text{Total Charges}_{it}}$
Structural Transfers	Siphon Ratio (SR_{it})	$\frac{\text{Home Office Adjustments}_{it}}{\text{Total Operating Costs}_{it}}$
Solvency	Current Ratio	$\frac{\text{Current Assets}_{it}}{\text{Current Liabilities}_{it}}$
Profitability	Return on Assets (ROA)	$\frac{\text{Net Income}_{it}}{\text{Total Assets}_{it}}$
Profitability	Operating Margin	$\frac{\text{Net Patient Revenue}_{it} - \text{Total Operating Expenses}_{it}}{\text{Net Patient Revenue}_{it}}$
Cost Structure	Labor Intensity	$\frac{\text{Total Salaries \& Benefits}_{it}}{\text{Total Operating Expenses}_{it}}$
Cash-Flow Dynamics	Accounts Receivable Days	$\frac{\text{Accounts Receivable}_{it}}{\text{Net Patient Revenue}_{it}/365}$
Cash-Flow Dynamics	Accounts Payable Days	$\frac{\text{Accounts Payable}_{it}}{\text{Total Operating Expenses}_{it}/365}$
Payer Mix	Charity Care Share	$\frac{\text{Charity Care Cost}_{it}}{\text{Total Operating Expenses}_{it}}$

Note: Current ratio, operating margin, and days cash on hand are used exclusively to construct the supervisory reference and are not included as anomaly signals.

Table B.11: Derived Analytical Statistics Used in MOSAIC

Statistic	Formula	Interpretation
Facility-level residual deviation ($Z_{\text{resid},imt}$)	$\frac{X_{imt} - \bar{X}_{im,8}}{\hat{\sigma}_{im,12}}$	Self-baseline deviation relative to trailing mean and volatility.
Temporal acceleration ($Z_{\text{resid_delta},imt}$)	$Z_{\text{resid},imt} - Z_{\text{resid},im,t-1}$	Change in self-baseline deviation capturing abrupt shifts.
Cross-regional divergence ($Z_{\delta,imt}$)	$\lvert Z_{\text{nat,signed},imt} - Z_{\text{reg,signed},imt}\rvert$	Difference between national and regional benchmark positions.
Mahalanobis distance (D_{it})	$\sqrt{(X_{it} - \mu_g)^\top \Sigma_g^{-1} (X_{it} - \mu_g)}$	Multivariate deviation from peer-cohort centroid.
Cross-measure dispersion ($\sigma_{\text{facility},it}$)	$\text{Std}_m(Z_{\text{peer,signed},imt})$	Within-facility inconsistency across measures.
Decoupling index (DCPL_{it})	$\frac{\text{Rev}_{it} - \text{Rev}_{i,t-1}}{\text{Rev}_{i,t-1}} - \frac{\text{Exp}_{it} - \text{Exp}_{i,t-1}}{\text{Exp}_{i,t-1}}$	Divergence between revenue and expense growth.
Operating-margin quadratic trend (CURV_{it})	$\hat{\beta}_2^{(i)}(\text{OM}_{i,t-5}, \ldots, \text{OM}_{it})$	Second-order operating-margin dynamics over a six-period window.
Residual volatility ($\sigma_{it}^{\text{resid},4}$)	$\text{Std}(Z_{\text{resid},i,t-3}, \ldots, Z_{\text{resid},it})$	Short-run volatility in baseline deviations.
Z-delta mean ($\bar{Z}_{\Delta,it}$)	$\frac{1}{\lvert M\rvert} \sum_{m \in M} \lvert Z_{\text{peer},imt} - Z_{\text{reg},imt}\rvert$	Average divergence between peer and regional positions.

Table B.12: Behavioral Anomaly Features Used in MOSAIC

Feature	Formula	Interpretation
Good performance magnitude (GoodZ_{imt})	$\max(0, Z_{\text{peer,signed},imt})$	Positive tail of peer-normalized performance.
Time since last elevated signal (TSLF_{it})	running count of $\mathbf{1}[\text{GoodZ}_{it}^{\text{comp}} > \tau]$	Duration of sustained above-threshold performance.
Robust peer score ($Z_{\text{peer},imt}^{\text{rob}}$)	$\frac{0.6745(X_{imt} - \text{median}_g(X_{mt}))}{\text{MAD}_g(X_{mt})}$	Robust peer-standardized deviation.
Peer-weighted GoodZ ($\text{GoodZ}_{\text{weighted},imt}$)	$\frac{\text{GoodZ}_{imt}}{\sigma_{\text{group},gmt}}$	Amplifies deviations in stable peer groups.
Bunching (BUNCH_{it})	$\frac{1}{\min_k \lvert \text{OM}_{it} - \tau_k \rvert + \varepsilon}$	Clustering near reporting thresholds.
Coordinated improvement (COORD_{it})	$\frac{1}{\lvert M \rvert} \sum_{m \in M} \mathbf{1}[Z_{\text{resid_delta},imt} > 0]$	Simultaneous improvement across measures.
Tradeoff signal (TRADE_{it})	$\max(0, \text{COORD}_{it}^{\text{targeted}} - \text{COORD}_{it}^{\text{untargeted}})$	Selective improvement in targeted metrics.
Peer divergence (PDIV_{it})	$\frac{1}{\lvert M \rvert} \sum_{m \in M} \lvert Z_{\text{peer}\times\text{year},imt} \rvert$	Average outlier magnitude across measures.
Rank instability (RINST_{it})	$\frac{1}{\lvert M \rvert} \sum_{m \in M} \lvert \text{Rank}_{im,t} - \text{Rank}_{im,t-1} \rvert$	Volatility in peer-relative ranking.

Table B.13: Model-Based Components in MOSAIC

Component	Role	Output Statistic
Variational Autoencoder (VAE)	Learned signal; MOSAIC	VAE_{it}
Isolation Forest (IF)	Density-based signal; MOSAIC and standalone comparator	IF_{it}
Transformer MAE	Masked autoencoder signal; MOSAIC aggregation and standalone	TR_{it}

Table B.14: Partial-Coverage Signals (Decile-Encoded) and Panel Coverage Rates

Signal	Coverage	Channel
Benford leading-digit divergence	37.9%	Model-based
Operating margin bunching signal	94.6%	Behavioral
Weighted GoodZ performance score	94.4%	Behavioral
Revenue–expense decoupling index	87.5%	Analytical
Operating-margin quadratic trend	63.5%	Analytical
Peer-rank instability (mean)	90.3%	Analytical
Peer divergence (year)	99.2%	Analytical
Rolling residual volatility (4-year)	56.5%	Analytical
Mean peer-state Z-delta	94.4%	Analytical

Appendix C. Robustness Analysis

Bootstrap Uncertainty Quantification

To assess the stability of learned signal weights, we apply a bootstrap perturbation procedure over 50 replicates. In each replicate, the $K \times n_{cr}$ signal matrix for each cohort–regime cell is row-resampled with replacement, and the full optimization is re-run to convergence. The realized weight distribution across replicates is summarized for each signal as the mean, standard deviation, and 10th–90th percentile interval of $w_k^{(c,r)}$.

For the five highest-weighted signals in the global solution, bootstrap standard deviations are small relative to the point estimates: the quality-deterioration missing-bin signal ($w = 0.064$) has $\hat{\sigma}_w = 0.003$; the quality-deterioration first-decile signal ($w = 0.062$) has $\hat{\sigma}_w = 0.004$. By contrast, the VAE reconstruction error signal ($w = 0.008$) exhibits $\hat{\sigma}_w = 0.006$, indicating that the optimizer's near-complete suppression of the structural channel is stable across bootstrap samples, not an artifact of a local minimum. The Spearman concordance ρ_S varies by less than ± 0.005 across all 50 replicates, confirming that the composite score is insensitive to sampling variation in the training panel.

Anchor Ablation

To assess sensitivity to the distress anchor specification, we re-optimize MOSAIC under three ablated anchor variants: (a) CR-only anchor ($d_{it} = a_{it}^{CR}$); (b) OM-only anchor ($d_{it} = a_{it}^{OM}$); (c) DCOH-only anchor ($d_{it} = a_{it}^{DCOH}$). In all three cases, the realized global Spearman concordance against the full three-component anchor exceeds $\rho_S = 0.51$, and the rank correlation between full-anchor MOSAIC scores and each ablated-anchor MOSAIC score exceeds 0.94. The dimensional weight allocation is qualitatively unchanged: behavioral dominance (47–55% of global weight) and latent structural suppression (VAE weight $< 1.5\%$) persist across all three ablation variants. These results confirm that the framework's behavioral preference is a property of the signal architecture rather than a consequence of the specific anchor parameterization.

Regime Boundary Sensitivity

The regime partition (Transition FY2011, Baseline FY2012–2018, Shock FY2019–2021, Recovery FY2022+) is researcher-defined. To assess sensitivity to this choice, we re-estimate MOSAIC under a three-regime alternative

collapsing Shock and Recovery into a single post-Baseline period (FY2019–2024). The global ρ_S varies by less than 0.008 from the four-regime specification, and the regime-specific concordance values for Transition and Baseline remain within 0.04 of those reported in Table 4. The COVID-shock years (FY2019–2021) remain the period of greatest single-model degradation in all specifications, confirming that the regime structure captures a genuine distributional break rather than a researcher degree of freedom.

Multi-Start Convergence Diagnostics

With $N_{\text{restart}} = 10$ initializations per cell as the operational setting, we assess whether the chosen solution is globally competitive by conducting an extended convergence diagnostic of 25 restarts on the global optimization problem (full panel). The five lowest objective values across those 25 diagnostic restarts cluster within 0.004 objective units of the reported solution ($\mathcal{L}^* = -0.5655$), suggesting the landscape has a well-defined basin around the global optimum and that $N_{\text{restart}} = 10$ is sufficient for production runs. For cohort–regime cells with $n_{cr} > 500$, the inter-restart objective range is typically below 0.01. For cells with $n_{cr} < 200$ (primarily critical access hospital shock-period cells), the inter-restart spread widens to 0.03–0.06, motivating the hierarchical shrinkage estimator that borrows from the global solution in precisely these cases.

Signal-Exclusion Robustness Check (MOSAIC-X)

To assess whether MOSAIC's concordance with the financial distress anchor is driven by the two signal families sharing limited informational lineage with the anchor's operating margin component, we re-estimated the full optimization pipeline after removing operating-margin bunching and operating-margin quadratic trend from the signal catalog. The resulting variant, MOSAIC-X, operates over 86 encoded signals rather than 108.

MOSAIC-X achieves $\rho_S = 0.5394$ ($\rho_{\text{peer}\times\text{yr}} = +0.5552$, separation $= 0.3923$), compared with $\rho_S = 0.5618$ for the full model—a reduction of 0.0224 despite the removal of signals carrying approximately 20% of global weight. The dimensional weight redistribution under MOSAIC-X shows the optimizer compensating primarily through increased loading on quality-deterioration and temporal trajectory signals, consistent with the behavioral dominance pattern observed in the full model. These results confirm that MOSAIC's discriminative power is a property of the full signal architecture rather than a consequence of anchor-feature overlap in the operating-margin channels.

Appendix D. Mathematical Programming: Complete Specification

This appendix provides the complete set-theoretic specification of the MOSAIC optimization problem. Section 2.6 in the main text presents the objective function and key constraints; the full notation is collected here for reference.

Sets

$i \in \mathcal{I}$	Hospitals (facility index)
$t \in \mathcal{T}$	Years (time index)
$k \in \mathcal{K}$	Anomaly-detection signals, $\lvert\mathcal{K}\rvert = K$
$c \in \mathcal{C}$	Peer cohorts (government, nonprofit, for-profit, other, CAH)
$r \in \mathcal{R}$	Time regimes (Transition, Baseline, COVID-Shock, Recovery)

The cohort assignment of hospital i is given by $c(i)$; regime assignment is given by $r(t)$. The cohort–regime cell containing observation (i, t) is $\mathcal{I}_{c,r} = \{(i, t) : c(i) = c,\ r(t) = r\}$.

Data

The following quantities are observed from HCRIS:

x_{itk}	Raw value of signal k for hospital i in year t
CR_{it}	Current ratio
OM_{it}	Operating margin
$DCOH_{it}$	Days cash on hand

The anchor variables (current ratio, operating margin, and days cash on hand) are selected for their consistent availability, clear economic interpretation, and established relationship to financial distress (Bazzoli et al., 2008). These variables are used exclusively to construct the supervisory reference d_{it} and do not enter the anomaly signal catalog.

Parameters

$\tilde{S}_{it}^{(k)} \in [0,1]$	Percentile-normalized anomaly signal
$K \in \mathbb{N}$	Total number of signals ($K = 108$)
$d_{it} \in [0,1]$	Supervisory reference constructed from $(CR, OM, DCOH)$
$\lambda_p \in [0,1]$	Hierarchical pooling regularization weight
α_c	Shrinkage weight based on Baseline sample size

Within each cohort c, cross-signal concordance is summarized by Kendall's $W^{(c)}$ computed over baseline observations. Soft-rank approximations are used in implementation to enable gradient-based optimization.

Decision Variables and Constraints

The decision variables $w_k^{(c)} \geq 0$ are defined at the cohort level and estimated exclusively using Baseline observations. They are subsequently held fixed and applied across all regimes.

The weights lie on the probability simplex:

$$\sum_{k=1}^{K} w_k^{(c)} = 1, \qquad w_k^{(c)} \geq 0 \quad \forall\, c \in \mathcal{C}. \tag{C1}$$

The separation statistic is defined over Baseline observations as

$$\Delta^{(c)}(\mathbf{w}) = \bar{F}_{\mathcal{H}_{c,\text{base}}} - \bar{F}_{\mathcal{L}_{c,\text{base}}}, \tag{C2}$$

where

$$\mathcal{H}_{c,\text{base}} = \{(i,t) : c(i) = c,\ d_{it} \geq 0.6,\ r(t) = \text{Baseline}\},$$
$$\mathcal{L}_{c,\text{base}} = \{(i,t) : c(i) = c,\ d_{it} \leq 0.3,$$
$$r(t) = \text{Baseline}\}.$$

Violations of monotonic ordering ($\Delta^{(c)} < 0$) are penalized within the objective function rather than imposed as a hard constraint.

Weight Parameterization

Weights are parameterized using a softmax transformation:

$$w_k^{(c)} = \frac{\exp(z_k^{(c)})}{\sum_{j=1}^{K} \exp(z_j^{(c)})},$$

where $z_k^{(c)} \in \mathbb{R}$. This guarantees nonnegativity and unit-sum constraints while enabling unconstrained optimization.

Optimization Procedure

For each cohort c:

1. Estimate global weights $\mathbf{w}_{\text{global}}$ on Baseline observations.
2. For cohorts with sufficient Baseline observations, solve the cohort-level problem using multi-start L-BFGS optimization.
3. Apply empirical Bayes shrinkage:

$$\mathbf{w}^{*(c)} = \alpha_c \, \hat{\mathbf{w}}^{(c)} + (1 - \alpha_c) \, \mathbf{w}_{\text{global}}, \qquad \alpha_c = \frac{n_c^{\text{Baseline}}}{n_c^{\text{Baseline}} + 50}.$$

4. Apply $\mathbf{w}^{*(c)}$ uniformly across all regimes.

Threshold Selection

The anomaly threshold τ^* is determined using a constrained optimization criterion:

$$\tau^* = \arg\max_{\tau} \left[\mathbb{E}(F_{it} \mid D_{it} = 1) - \mathbb{E}(F_{it} \mid D_{it} = 0) \right]$$

subject to

$$\Pr(F_{it} \geq \tau \mid D_{it} = 0) \leq \alpha, \quad \alpha = 0.05,$$

where $D_{it} = \mathbf{1}(d_{it} \geq 0.6)$.

Coefficient Justification

The concordance weighting coefficient ($\frac{1}{4}$) ensures that cross-signal agreement contributes to the objective without dominating alignment with the supervisory reference. The monotonicity penalty coefficient (5) enforces directional identification by penalizing violations of the separation condition. These values were selected through empirical sensitivity analysis to balance stability and identification strength across heterogeneous cohorts.

Appendix E. Operational Deployment Guide

This appendix provides guidance for regulatory users and institutional researchers seeking to deploy the MOSAIC framework for audit prioritization or adapt it to analogous administrative data contexts.

E.1 Score Interpretation

MOSAIC scores should be interpreted as *ordered risk ranks*, not as probability or magnitude estimates. Score compression (mean = 0.347, SD = 0.237) is a structural consequence of the simplex constraint averaging over 108 signals; the absolute distance between two scores carries no direct inferential content. The operationally meaningful quantities are: (a) rank within the relevant cohort–regime cell for within-cohort prioritization; (b) the assigned risk category (Table 5) for across-cohort triage; and (c) the per-dimension weight decomposition for investigation guidance.

E.2 Per-Facility Score Decomposition

For each flagged facility-year (i, t), the contribution of analytic dimension d to the MOSAIC composite is

$$\mathrm{Contrib}_{it}^{(d)} = \sum_{k \in \mathcal{K}_d} w_k^{*(c(i),r(t))}\, \tilde{S}_{it}^{(k)},$$

where $\mathcal{K}_d$ is the set of signals assigned to dimension d in the signal catalog (Appendix A). Summing across dimensions recovers the full MOSAIC score: $\sum_d \mathrm{Contrib}_{it}^{(d)} = \mathrm{MOSAIC}_{it}$. This decomposition is computable directly from the MOSAIC weight matrix and the normalized signal matrix, requiring no additional estimation. For audit-report integration, we recommend reporting dimensional contributions as a stacked bar or radar chart alongside the composite score and risk category, enabling auditors to identify which anomaly dimension is driving the flag before selecting the appropriate investigative protocol.

E.3 Audit Prioritization Workflow

A recommended triage workflow proceeds in four steps. **Step 1**: Apply the Bastian threshold τ^* (or an institution-specific FPR target) to identify the flagged population. At $\alpha = 0.05$, this yields approximately 10.4% of facility-years ($n \approx$ 8,689 at current panel scale). **Step 2**: Within the flagged population, sort by risk category (Critical first, then Structural+Behavioral, then Behavioral+Statistical). The 188 Critical-flagged facility-years represent the highest-priority subset for immediate forensic review. **Step 3**: Within each risk category, sort by cohort–regime rank rather than raw MOSAIC score to control for cohort-level heterogeneity in absolute score distribution. **Step 4**: For each facility selected for investigation, retrieve the per-dimension score

decomposition (Step E.2) and the top contributing signal identifiers from the signal catalog (Appendix A) to guide document requests and on-site review protocols.

E.4 Recalibration Schedule

The MOSAIC framework embeds four time-sensitive components that require periodic recalibration. (i) *Decile cut-points*: estimated from the 2012–2018 Baseline period and frozen. As the Baseline period ages, cut-points may drift from the current distribution; we recommend re-estimating cut-points every five years or following any major revision to HCRIS field definitions. (ii) *Peer-group scaling parameters* (median and IQR within cohort–regime cells): re-estimated annually as new fiscal-year data become available. (iii) *Signal weights*: re-optimized annually after the full panel update. The hierarchical shrinkage structure means that small cells update conservatively toward the global estimate, limiting overfitting to single-year fluctuations. (iv) *Bastian threshold* τ^*: re-estimated annually from the updated panel; shifts in the threshold value across years can themselves serve as a diagnostic for structural change in the anomaly distribution.

E.5 Portability to Other Administrative Datasets

The MOSAIC architecture is not specific to HCRIS. The components that are portable without modification are: the 11-bin decile encoding scheme for partial-coverage signals; the multi-objective optimization formulation (Appendix D); the hierarchical shrinkage estimator; and the Bastian auto-threshold rule. Components requiring domain-specific adaptation are: the signal catalog (Appendix A), which must be rebuilt from the target dataset's field structure; the peer cohort definitions, which must reflect the institutional taxonomy of the new domain; and the distress anchor, which must be constructed from outcome-adjacent financial or operational variables that are consistently reported and free of direct algebraic overlap with the signal catalog. The decile encoding and optimization framework have been applied without modification to skilled nursing facility (SNF) cost reports in ongoing work, suggesting broad portability within the CMS administrative data ecosystem.

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